



### MPP- 611AL

#### Introduction

MPP-611 is one of Micro Powders' most popular additives for coatings requiring excellent rub and mar resistance with good surface slip, and anti-blocking properties. MPP-611AL is a micronized composite based on high density polyethylene reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9.).

| Property:                  | MPP-611XF | MPP-611AL |
|----------------------------|-----------|-----------|
| Mean particle size (µm)    | 4.0 – 6.0 | 3.5 – 5.5 |
| Maximum particle size (µm) | 22.0      | 15.56     |
| Melting point (°C)         | 110 - 116 | 110 – 116 |
| Density at 25 °C (g/cc)    | 0.96      | 0.99      |

#### Objective

Compare the level of scratch resistance between MPP-611XF and MPP-611AL.

#### Study Methodology

Pencil scratch tests were conducted on a water based polyurethane coating:

| Formula:                          | Weight %: |
|-----------------------------------|-----------|
| UROTUF® L522-W-40 (Reichhold)     | 89.6      |
| Diethylene glycol monoethyl ether | 5.0       |
| BYK-345                           | 0.5       |
| TEGO® Foamex 805 N                | 0.3       |
| Water                             | 4.6       |
| Total:                            | 100.0     |
| Nonvolatiles (Solids):            | 35.7      |

The polyurethane coating was dosed with 1% of MPP-611XF and MPP-611AL. Coatings were drawn down with a #28 wire wound rod to produce approximately 1 mil DFT coatings. Coatings were dried for 7 days at room temperature and ~50% humidity.

The pencil scratch hardness test was conducted under controlled measurement conditions using a Taber Linear Abraser.

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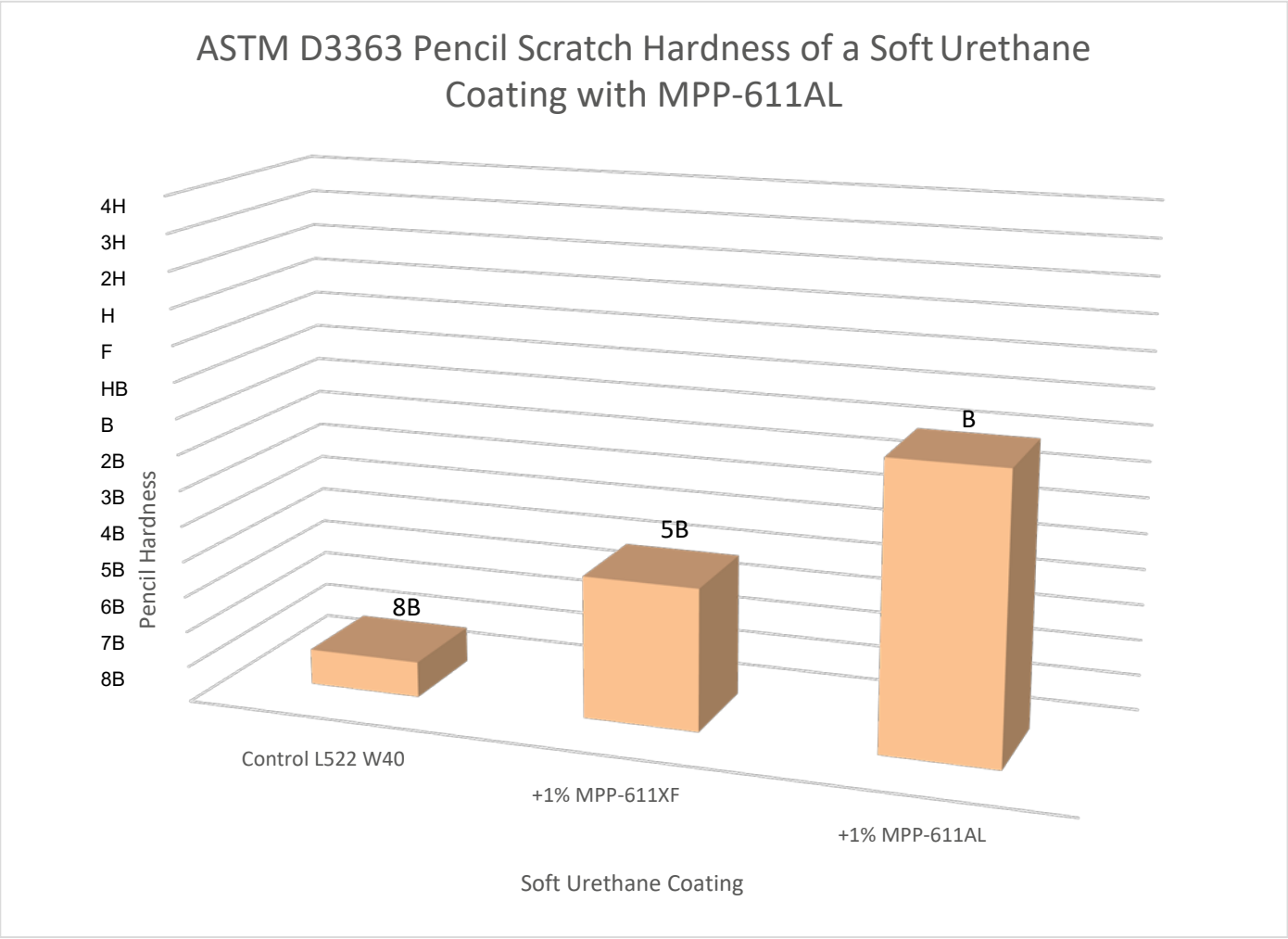
### **Taber Linear Abraser – Pencil Scratch Test Instrument**

The motion of the abrasion is linear - one cycle is equal to one forward and one backward stroke of the test arm. The Taber Linear Abraser 5750, coupled with the Pencil Hardness kit accessory, enables controlled and reproducible Pencil Scratch Hardness testing of coatings by allowing the user to control the amount of force applied during the measurement, the number of linear cycles, the stroke length, as well as the speed. For this particular study, a force of 5.3 Newtons (equivalent to 536 g) was applied during the stroke. The stroke length was set to 0.5 inches and the speed was set to 2 cycles per minute. Each coating was scratched for one-half of a cycle (in the forward motion, as specified in ASTM D3363). These specific test conditions allowed one to observe differences in the amount of scratch for each dosed and undosed coating. Pencil scratch was assessed visually by comparing the surfaces of the scratched coatings and assigning the scratch hardness rating.



Results

| Pencil Scratch Hardness Rating |    |    |    |    |    |                                |   |      |   |   |    |    |    |
|--------------------------------|----|----|----|----|----|--------------------------------|---|------|---|---|----|----|----|
| Soft                           |    |    |    |    |    | Mid-Point                      |   | Hard |   |   |    |    |    |
| 8B                             | 7B | 6B | 5B | 4B | 3B | 2B                             | B | HB   | F | H | 2H | 3H | 4H |
| Test Samples                   |    |    |    |    |    | Pencil Scratch Hardness Rating |   |      |   |   |    |    |    |
|                                |    |    |    |    |    | 1 MIL DFT                      |   |      |   |   |    |    |    |
| Control (No Additive)          |    |    |    |    |    | 8B                             |   |      |   |   |    |    |    |
| + 1% MPP 611XF                 |    |    |    |    |    | 5B                             |   |      |   |   |    |    |    |
| + 1% MPP 611AL                 |    |    |    |    |    | B                              |   |      |   |   |    |    |    |



## Conclusions

- MPP-611AL provides a dramatic improvement in Taber linear abrasion (pencil scratch hardness)
- MPP-611AL gives the formulator the cutting-edge performance of alumina nanopowder in a safe, easier-to-disperse, non-nano additive
- MPP-611AL has a density optimized for maximum surface efficiency and minimal settling vs. alumina nanopowders